Pseudocodes

Q1

```
START
DECLARE seats[50][50]
DECLARE rows, columns, k
DECLARE i, j, count, available, total
DECLARE middle, distance, side, start, found

INPUT rows and columns
WHILE rows<1 OR rows>50 OR columns<1 OR columns>50
   DISPLAY "Invalid input"
   INPUT rows and columns
ENDWHILE

FOR i=0 TO rows-1
    FOR j=0 TO columns-1
       INPUT seat value

       WHILE seat value is<>0 AND seat value<>1
          DISPLAY "Invalid seat value"
          INPUT seat value
       ENDWHILE

       SET seats[i][j]=seat value
   ENDFOR
ENDFOR

DISPLAY "Seating Arrangement"
FOR i=0 TO rows-1
   FOR j=0 TO columns-1
      IF seats[i][j]==0
         DISPLAY "_"
      ELSE
         DISPLAY "X"
      ENDIF
   ENDFOR
```

```
    Move to Next line
ENDFOR
SET total=0

FOR i=0 TO Rows-1
    SET availabl=0
    FOR j=0 TO Columns-1
        IF seats[i][j]==0
            INCREASE available By 1
        ENDIF
    ENDFOR
    DISPLAY available seats in current Row
    ADD available TO total
ENDFOR
DISPLAY total available seats
INPUT number of consecutives seats requseed (k)

WHILE K<1 OR K>columns
    DISPLAY "Invalid number of seats"
    INPUT k
ENDWHILE
SET middle=Rows/2
SET found=0

FOR distance=0 TO Rows-1
    FOR side=0 TO 1
        IF side==0
            SET i=middle-distance
        ELSE
            SET i=middle+distance
        ENDIF
```

```
        IF i>=0 AND i<rows
          SET count=0
          FOR j=0 TO columns-1
            IF seats[i][j]==0
              INCREASE count by 1
            ELSE
              SET count=0
            ENDIF
            IF count==k
              SET start=j-k+1
              SET found=1
              EXIT search
            ENDIF
          ENDFOR
        ENDIF
    ENDFOR
ENDFOR
        IF found==1
          FOR j=start TO start+k-1
            SET seats[i][j]=1
          ENDFOR
        DISPLAY "seats booked successfully"
        SUBTRACT k from total
        DISPLAY updated seating arrangement
        DISPLAY remaining available seats
        ELSE
          DISPLAY "Required consecutive seats are not available"
        ENDIF

        END
```

Q2

```
START
DECLARE n, i, second, count
INPUT n
WHILE n<=0
    DISPLAY "Enter a positive number"
    INPUT n
ENDWHILE

SET count=0
FOR i=2 TO n
   SET sum1=0
   FOR j=1 TO i/2
      IF i MOD j==0
         ADD j TO sum1
      END IF
   END FOR

SET second=sum1
IF seconds>i AND second<=n
   SET sum2=0
   FOR j=1 TO second/2
      IF second>i AND second<=n
         SET sum2=0
         FOR j=1 TO second/2
            IF second MOD j==0
               ADD j TO sum2
            END IF
   END FOR

   IF sum2==i
      DISPLAY i and second
      INCREASE count BY 1
   END IF
ENDFOR ENDIF
```

```
IF count == 0
   DISPLAY "No amicable pairs Found"
ELSE
   DISPLAY total number of amicable pairs
ENDIF
END
START FUNCTION sumDivisors(num)
DECLARE sum=0
FOR i=1 TO num/2
   IF num MOD i==0
      ADD i TO sum
   ENDIF
ENDFOR
RETURN sum
END FUNCTION
```

Q3

```
START
DECLARE temperature[50][50]
DECLARE sensors, hours
DECLARE i,j
DECLARE total, average, highest, lowest
DECLARE critical, warning, streak
DECLARE highestAverage, highestSensor

CALL InputReadings()
FOR i=0 TO sensors-1
   CALL analyzeSensor(i)
ENDFOR

CALL findHighestAverage()
DISPLAY greenhouse temperature report
END
```

```
FUNCTION inputReadings()
    INPUT number of sensors
    INPUT number of readings
    FOR i=0 TO sensors-1
        FOR j=0 TO hours-1
            INPUT temperature reading
            STORE reading in temperature[i][j]
        ENDFOR
    ENDFOR
ENDFUNCTION

FUNCTION analyzeSensor(i)
    SET total=0
    SET critical=0
    SET warning=0
    SET streak=0
    SET highest=first reading of current sensor
    SET lowest=first reading of current sensor

    FOR j=0 TO hours-1
        ADD temperature[i][j] TO total
        IF temperature[i][j] > highest
            SET highest=temperature[i][j]
        ENDIF
        IF temperature[i][j] < lowest
            SET lowest=temperature[i][j]
        ENDIF
        IF temperature[i][j] >= 40
            INCREASE critical BY 1
        ELSE IF temperature[i][j] >= 35
            INCREASE warning BY 1
        ENDIF
```

CamScanner

```
        IF temperature[i][j] >= 35
            INCREASE streak BY 1
        ELSE
            SET streak = 0
        ENDIF
    ENDFOR

    DISPLAY total temperature
    DISPLAY average temperature
    DISPLAY highest temperature
    DISPLAY lowest temperature
    DISPLAY critical readings
    DISPLAY warning readings
    DISPLAY consecutive high-temperature readings

    STORE average for current sensor
    ENDFUNCTION

    FUNCTION findHighestAverage()
        SET highestAverage = average of first sensor
        SET highestSensor = first sensor
        FOR i = 1 TO sensors - 1
            IF sensor average > highestAverage
                SET highestAverage = sensor average
                SET highestSensor = i
            ENDIF
        ENDFOR

        DISPLAY sensor with highest average temperature
        ENDFUNCTION
```